

OpenClaw API-KEY Configuration

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1. Course Overview

This course will detail how to configure the AI model API-KEY for OpenClaw, including the application process for API-KEYs, two configuration methods (CLI command configuration and configuration file modification), and how to verify if the configuration is successful. After completing this course, you will be able to successfully connect to and use various AI large model service providers.

1.1 Course Content

- Understand the types of model service providers supported by OpenClaw
- Master the method for applying for API-KEYs
- Learn how to view and verify the status of connected models

2. API-KEY Application

2.1 Recommended Model Service Providers

- **Recommended for beginners:** Refer to the accompanying tutorial [Chapter 8 Online AI Large Model Development] — [Online Large Models (Text Interaction)] — [2. Tongyi Qianwen Large Model] to apply for the corresponding model provider API-KEY.
- OpenClaw supports numerous model service providers, including:
 - **OpenAI** - GPT series (API + Codex)
 - **Anthropic** - Claude series (API + Claude Code)
 - **Google** - Gemini series

- **Alibaba Cloud** - Qwen Cloud, Tongyi Qianwen
- **DeepSeek** - DeepSeek models
- **Zhipu AI** - GLM models
- **Moonshot AI** - Kimi + Kimi Coding
- **Ollama** - Local models (cloud + local)
- And many other service providers
- You can freely choose a model service provider based on your actual needs and regional network policies, register on the corresponding official website, and apply for an API-KEY.

3. Configuring OpenClaw API-KEY

3.1 Configuration Methods

OpenClaw provides two ways to configure API-KEYs:

1. **CLI Command Configuration** (suitable for beginners)
2. **Directly Modify Configuration File** (suitable for developers)
 - Configuration file path: `$HOME/.openc1aw/openc1aw.json`

3.2 Quick Configuration

- Suitable for users who want to get started quickly. You need to register for an Alibaba Cloud Bailian large model account and create an API-KEY first (refer to the accompanying tutorial [Chapter 8 Online AI Large Model Development] — [Online Large Models (Text Interaction)] — [2. Tongyi Qianwen Large Model]).
- The factory-preset configuration file has already configured the Bailian model parameters; just fill in the API-KEY to use it directly.
- Open the OpenClaw configuration file:

```
nano $HOME/.openc1aw/openc1aw.json
```

```
nano $HOME/.openc1aw/agents/main/agent/models.json
```

3.2.1 Configure baseUrl

- Overseas users should select the corresponding baseUrl according to their service region (go to [ModelStudio Console](#) homepage, switch to the target region in the upper right corner, such as **Singapore**, then select **API Key** in the left navigation bar to create an API Key)

Base URL for the region where the API key belongs:
 Singapore: `https://dashscope-intl.aliyuncs.com/compatible-mode/v1`
 US (Virginia): `https://dashscope-us.aliyuncs.com/compatible-mode/v1`

3.2.2 Fill in apiKey

```
"models": {
  "mode": "merge",
  "providers": {
    "bailian": {
      "baseUrl": "https://dashscope.aliyuncs.com/compatible-mode/v1",
      "apiKey": "sk-a87e1137b17b4a35af13aa4a12ac3473",
```

- After filling it in, press `Ctrl+X`, then press `Y` to confirm saving, and press Enter to exit.

4. View Connected Model Status

4.1 View the List of Configured Models

After configuration, you can use the following command to view the connected models:

```
openclaw models list
```

```
sunrise@ubuntu:~/openclaw$ openclaw models list
🔔 OpenClaw 2026.3.24 (cff6dc9) – I don't have opinions about tabs vs spaces. I have opinions about everything else.
Model      Input  Ctx  Local Auth  Tags
deepseek/deepseek-v4-pro  text  977k  no   yes  default
```

4.2 Verify Configuration Success

1. Test Conversation

```
openclaw tui
```

Start the TUI interface, send a test message (e.g., "hello"), and confirm that the model responds normally.

```
May I have your name

My name is 小亚 (Xiao Ya).

I'm your robot assistant, running on OpenClaw with a physical body equipped
with servos, sensors, RGB lights, OLED display, and a camera. I'm here to help
you with hardware control, vision tasks, and various smart applications.

Nice to meet you! 🤖
connected | idle
agent main | session main (openclaw-tui) | bailian/qwen3.5-plus | tokens
41k/1.0m (4%)
```

5. Other Access Methods (Advanced)

- If you have your own model provider or need to access other model providers later, refer to the following tutorial.

5.1 CLI Command Configuration for API-KEY (Suitable for Beginners)

1. Start Configuration Wizard

```
openclaw onboard
```

- Select [Yes] and press Enter.

```
Recommended baseline:
- Pairing/allowlists + mention gating.
- Multi-user/shared inbox: split trust boundaries (separate
  gateway/credentials, ideally separate OS users/hosts).
- Sandbox + least-privilege tools.
- Shared inboxes: isolate DM sessions (`session.dmScope:
  per-channel-peer`) and keep tool access minimal.
- Keep secrets out of the agent's reachable filesystem.
- Use the strongest available model for any bot with tools or
  untrusted inboxes.
```

```
Run regularly:
openclaw security audit --deep
openclaw security audit --fix
```

```
Must read: https://docs.openclaw.ai/gateway/security
```

```
◆ I understand this is personal-by-default and shared/multi-user use requires
lock-down. Continue?
  ● Yes / ○ No
```

- Select QuickStart for the configuration mode and press Enter.

```
◆ Setup mode
  ● QuickStart (Configure details later via openclaw configure.)
  ○ Manual
```

2. Select Model Service Provider

- Use the up and down arrows to select the service provider for which you have applied for an API-KEY, and press Enter to confirm. Here we take Alibaba Cloud Bailian model provider as an example; the configuration method for other model providers is the same.
- Here select Custom Provider **[Custom Provider]**

```
◆ Model/auth provider

Search:
  ○ Alibaba Model Studio
  ○ Anthropic
  ○ Arcee AI
  ○ BytePlus
  ○ Chutes
  ○ Cloudflare AI Gateway
  ○ Copilot
  ● Custom Provider (Any OpenAI or Anthropic compatible endpoint)
```

- Enter the base_url of the model provider.

```
◆ API Base URL
  https://dashscope.aliyuncs.com/compatible-mode/v1
```

3. Enter API-KEY

- After selecting the provider, the system will prompt you to enter the corresponding API-KEY.

```
◆ How do you want to provide this API key?  
● Paste API key now (Stores the key directly in OpenClaw config)  
○ Use external secret provider
```

- Paste or type the API-KEY you applied for, and press Enter to confirm.

```
◆ API Key (leave blank if not required)  
sk-a87e1137b17b4a35af13aa4a1xxxxxx
```

- Select **OpenAI-compatible** for the interface protocol.

```
◆ Endpoint compatibility  
● OpenAI-compatible (Uses /chat/completions)  
○ Anthropic-compatible  
○ Unknown (detect automatically)
```

4. Fill in the model name, taking MiniMax as an example.

```
◆ Model ID  
MiniMax-M2.5
```

- **Endpoint ID** is the model provider identifier and can be filled in freely.

```
◆ Endpoint ID  
custom-dashscope-aliyuncs-com
```

- Model alias (optional) can be left blank.

```
◆ Model alias (optional)  
e.g. local, ollama
```

5. Other Configurations

- For other configuration items, select **Skip** for all as shown below; no configuration is needed.

```
◇ Skills status

Eligible: 16
Missing requirements: 38
Unsupported on this OS: 7
Blocked by allowlist: 0

◆ Configure skills now? (recommended)
  ○ Yes / ● No
```

- For Enable Hooks, press space to select **Skip for now**.

```
Enable hooks?
■ Skip for now
□ 🚀 boot-md
□ 📎 bootstrap-extra-files
□ 📝 command-logger
□ 💾 session-memory
```

- Select Restart to restart the gateway and apply the configuration.

```
Gateway service already installed
● Restart
○ Reinstall
○ Skip
```

- Close the popped-up OpenClaw webpage.

5.2 Modify Configuration File (Recommended Method)

Configuration File Location

Configuration files are located at: `$HOME/.openclaw/openclaw.json`

```
`$HOME/.openclaw/agents/main/agent/models.json`
```

Edit Configuration File

1. Open Configuration File 1

```
nano $HOME/.openclaw/openclaw.json
# Or use your preferred editor, such as vim, code, etc.
```

2. Add Model Provider Configuration

Add or modify the `models.providers` section in the configuration file:

```
{
  "models": {
    "providers": {
      "openai": {
        "apiKey": "OPENAI_API_KEY"
      },
      "anthropic": {
        "apiKey": "ANTHROPIC_API_KEY"
      },
      "google": {
        "apiKey": "GOOGLE_API_KEY"
      }
    }
  },
  "agents": {
    "defaults": {
      "model": {
        "primary": "openai/gpt-4"
      }
    }
  }
}
```

3. Open Configuration File 2

```
nano $HOME/.openclaw/agents/main/agent/models.json
```

4. Add Model Provider Configuration

Add or modify as follows:

```
"bailian": {
  "baseUrl": "https://dashscope.aliyuncs.com/compatible-mode/v1",
  "apiKey": "Your API Key",
  "api": "openai-completions",
  "models": [
    {
      "id": "qwen3.5-plus",
      "name": "qwen3.5-plus",
      "reasoning": false,
      "input": [
        "text",
        "image"
      ],
      "cost": {
        "input": 0,
        "output": 0,
        "cacheRead": 0,
        "cacheWrite": 0
      },
      "contextWindow": 1000000,
      "maxTokens": 65536,
      "api": "openai-completions"
    }
  ]
}
```

5. Configuration Description

- `models.providers`: Configure the API-KEYs for each model service provider.
 - `openai.apiKey`: OpenAI's API-KEY

- `anthropic.apikey`: Anthropic's API-KEY
- `google.apikey`: Google's API-KEY
- `agents.defaults.model.primary`: Set the default model to use.
 - Format is `provider/model_name`, e.g., `openai/gpt-4`, `anthropic/claude-3`, `google/gemini-pro`

6. Save and Exit

- Be careful not to randomly delete periods or add other symbols when modifying.
- Press `Ctrl+O` to save, `Ctrl+X` to exit (nano editor).

7. Factory Default Bailian Configuration (Can be directly modified or referenced)

```
"models": {
  "mode": "merge",
  "providers": {
    "bailian": {
      "baseUrl": "https://dashscope.aliyuncs.com/compatible-mode/v1",
      "apiKey": "sk-a98b519b6bab4e35a7b82b4a95b8a4f6",
      "api": "openai-completions",
      "models": [
        {
          "id": "qwen3.5-plus",
          "name": "qwen3.5-plus",
          "reasoning": false,
          "input": [
            "text",
            "image"
          ],
        },
        {
          "id": "MiniMax-M2.5",
          "name": "MiniMax-M2.5",
          "reasoning": false,
          "input": [
            "text"
          ],
        },
        {
          "id": "glm-4.7",
          "name": "glm-4.7",
          "reasoning": false,
```

```
    "input": [
      "text"
    ],
    "cost": {
      "input": 0,
      "output": 0,
      "cacheRead": 0,
      "cachewrite": 0
    },
    "contextwindow": 202752,
    "maxTokens": 16384,
    "compat": {
      "thinkingFormat": "qwen"
    }
  }
}
```

6. Common Issues and Solutions

6.1 Invalid API-KEY or Errors

Issue: Prompt indicates invalid API-KEY or authentication failure after configuration.

Solution:

1. Confirm that the API-KEY is correct (ensure there are no extra spaces).
2. Check if the API-KEY has expired or been disabled.
3. Confirm that the account balance is sufficient (some providers require pre-payment).

6.2 Unable to Access Model Service

Issue: Configuration is correct but unable to connect to the model service.

Solution:

1. Check if the network connection is normal.
2. Confirm whether the network in your region can access the provider (some services may require special network configuration).
3. Try switching to another model provider for testing.

7. Course Summary

Through this course, you have mastered:

- ☑ Types of model service providers supported by OpenClaw
- ☑ Application methods and precautions for API-KEYs
- ☑ CLI command configuration method (suitable for beginners)
- ☑ Configuration file modification method (suitable for developers)
- ☑ Methods to view and verify model configuration status
- ☑ Troubleshooting and solutions for common issues

